



Enhancing recommendation systems with collaborative filtering and sentiment analysis: dimensionality reduction for improved content-based approaches

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Abstract

A recommendation system plays a pivotal role in assisting users by providing personalized suggestions based on their preferences. However, these systems often neglect the crucial aspect of user sentiment. To address this limitation, this work focuses on integrating sentiment analysis into recommendation systems. Using a dataset of 5000 reviews from Yelp, we employ preprocessing tasks such as Stop Words Removal, Tokenization, Part-of-Speech Tagging, WordNet Part-of-Speech Information, and Lemmatization to facilitate sentiment analysis. Leveraging the power of BERT, our sentiment analysis model achieves an impressive accuracy rate of 91%. We then incorporate the sentiment analysis component into collaborative filtering, utilizing cosine similarity and significantly reducing the root mean square error (RMSE) to 1.2478. Additionally, we integrate sentiment analysis into content-based recommendation using clustering, resulting in improved recommendations with a higher silhouette score of 0.2270. To further enhance the system's performance, we propose a novel approach that combines these sentiment-aware components using NMF with DecisionTreeRegressor, resulting in an even lower RMSE of 1.1955. This integration of sentiment analysis into the recommendation system demonstrates its effectiveness in improving accuracy and personal-

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ization, providing users with more meaningful and relevant recommendations based on their sentiments.

Keywords Recommendation system · Sentiment analysis · BERT · Collaborative filtering · Content-based recommendation · Preprocessing techniques

1 Introduction

In the era of digital information, users are inundated with vast amounts of data from various online sources. This deluge of information, including user reviews on platforms such as Yelp, TripAdvisor, and Amazon, presents a significant challenge: how to effectively filter and recommend relevant content amidst the overwhelming choices available. Reviews, which can be positive, negative, or neutral, provide crucial insights into products, services, and experiences, but their sheer volume and variability make it difficult to extract meaningful recommendations [1–3].

The significance of this challenge lies in the impact that accurate recommendation systems have on user satisfaction and decision-making. Traditional recommendation systems, which primarily rely on collaborative filtering (CF) and content-based (CB) approaches, often struggle to capture the nuanced sentiments expressed in reviews [4]. Collaborative filtering compares user preferences with those of similar users, while content-based filtering focuses on item attributes. Although hybrid methods combine these approaches to improve accuracy, they may still fall short in fully leveraging the sentiment conveyed in user reviews [5–8]. Our proposed solution addresses these limitations by integrating advanced sentiment analysis techniques into the recommendation system. By employing BERT for sentiment analysis, we enhance the system's ability to interpret and utilize user sentiments, thereby improving the relevance and accuracy of recommendations. This hybrid approach combines BERT with collaborative filtering and content-based techniques, offering a more comprehensive and personalized recommendation experience.

Recommendation systems employ a variety of technologies to provide personalized suggestions to users. Some of the main technologies used in recommendation systems include:

- **Text Mining:** Text mining techniques are used to extract and analyze textual information from various sources, such as product descriptions, reviews, and user-generated content. This helps in understanding user preferences and item characteristics [9].
- **K-Nearest Neighbors (KNN):** KNN uses historical interactions to identify similar items or users, making it a collaborative filtering technique [10].
- **Clustering:** Clustering algorithms group similar items or users together. This can be useful in content-based recommendation systems, where items are clustered based on their features, and users are recommended items from the same cluster [11, 12].
- **Matrix Factorization:** A method that decomposes a user–item matrix into two lower-dimensional matrices, which can then be used to recommend items based on the similarity of their corresponding columns or rows [13, 14].
- **Neural Network:** Neural networks, especially deep learning models, are increasingly utilized in recommendation systems to capture intricate patterns in user behavior and item characteristics [11].

Sentiment analysis (SA), a domain in natural language processing, predicts the positive or negative polarity of an entity, providing insights into user attitudes, opinions, and emotions. It extracts valuable information from reviews, including overall sentiment toward a product

or service, specific liked or disliked features, and sentiment trends for different aspects of an item [15].

1. *What role does Sentiment Analysis play in recommendation systems?*

Sentiment Analysis (SA) is a crucial component that significantly enhances recommendation systems. By extracting and analyzing sentiments expressed in user reviews, sentiment analysis contributes in the following ways [16]:

- **Understanding User Preferences:** Sentiment analysis helps systems comprehend user sentiments toward specific attributes or experiences. This deep understanding enables recommendation systems to cater to individual preferences more effectively.
- **Improving Recommendations:** By considering both positive and negative sentiments, recommendation systems can suggest options that align with a user's emotional and experiential requirements, thereby increasing the likelihood of user satisfaction.
- **Enhancing Diversity:** Sentiment analysis promotes diversity in recommendations by capturing nuanced preferences of users, resulting in a more varied and personalized experience.

2. *Contributions:*

The novelty and contributions of our work can be summarized as follows:

- **Innovative Integration of Sentiment Analysis in Hybrid Recommendation Systems:** We introduce a new way to integrate sentiment analysis into a hybrid recommendation system using the BERT-base-uncased model with AdamW optimization. This approach improves our understanding of user sentiment, leading to better accuracy and personalization in recommendations, an area often neglected in existing systems.
- **Optimal Selection of Hyperparameters for BERT in Sentiment Analysis:** Through extensive experimentation with various hyperparameters, we determine the optimal settings for BERT, positioning it as the most effective model for sentiment analysis compared to traditional methods like Vader, TextBlob, and AFINN. This strategic choice underscores our commitment to high-performance models, resulting in enhanced sentiment interpretation that enriches the recommendation process.
- **Effective Use of Cosine Similarity in Collaborative Filtering with Sentiment Analysis Integration:** We show that cosine similarity is an effective metric for collaborative filtering, particularly when combined with sentiment analysis. By normalizing sentiment scores and incorporating them into the user-item interaction matrix alongside raw ratings, our model effectively utilizes both user ratings and sentiment data. We further validate this integration through a statistical t-test and employ fivefold cross-validation to ensure robust results. This integration significantly improves the accuracy of personalized recommendations, outperforming traditional methods.
- **Enhanced Content-Based Recommendation through Clustering:** We implement advanced clustering techniques within the content-based recommendation framework, effectively grouping items based on textual attributes using Word2Vec (Skip-gram) and normalizing numerical features with StandardScaler. To evaluate the effectiveness of different clustering algorithms, we tested a variety of methods and assessed their performance using the silhouette score. This evaluation allows us to select the most effective clustering approach, capturing item similarities with greater precision and leading to highly personalized and contextually relevant recommendations.
- **Development of a Comprehensive Hybrid Recommendation System:** We propose an advanced hybrid recommendation system that integrates collaborative filtering

(CF), content-based (CB) techniques, and sentiment analysis (SA) using a novel methodology. By combining nonnegative matrix factorization (NMF) and Decision-TreeRegressor through a feature combination strategy, our system synthesizes outputs from CF, CB, and SA to enhance recommendation scores. This approach improves the diversity, accuracy, and personalization of recommendations, providing a more effective solution than traditional models.

These contributions collectively enhance our recommendation system by incorporating SA, utilizing advanced models like BERT, and employing hybrid approaches that combine CF, CB filtering, and SA techniques.

3. *Paper Organization:*

The paper is organized into five sections: introduction, literature review, proposed system, results and discussion, and conclusion. The introduction sets the context and research problem, while the literature review explores existing research in the field. The proposed system section details the methodology and techniques used. The results and discussion section presents the experimental findings and their analysis. The conclusion summarizes the key findings and discusses potential future directions. Finally, the references section provides a list of cited sources.

2 Literature review

Adding sentiment analysis (SA) to recommendation systems (RS) is crucial as it captures emotional tone, leading to more personalized recommendations and reducing information overload by filtering content based on user sentiment. Additionally, it enhances trust by identifying biased reviews and improves overall system performance, resulting in better accuracy, reliability, and user satisfaction [17, 18].

The study [19] introduces a novel hybrid model that integrates ResNeXt with a recurrent neural framework (NH-ResNeXt-RNF) to enhance multi-class sentiment classification. This approach addresses the challenges associated with traditional machine learning and sentiment-based lexical methods, particularly for domain-specific problems. By employing unsupervised learning and preprocessing to eliminate noisy data, the model improves the efficiency of multi-class sentiment analysis and reduces the effort required for data annotation. The framework is evaluated on Amazon and Twitter datasets, demonstrating its potential to outperform conventional supervised, semi-supervised, and unsupervised sentiment analysis approaches.

The study [20] addresses the limitations of traditional binary or tri-classification approaches by introducing a novel heuristic bidirectional-recurrent neural network (NHBi-RNN) for multi-class sentiment classification. This approach overcomes challenges related to textual variations and complex logic by providing a more nuanced understanding of customer sentiments. The proposed framework includes comprehensive steps for data acquisition, preprocessing, feature extraction, and sentiment classification, enabling it to categorize feedback into five distinct sentiment classes: very positive, positive, neutral, negative, and very negative. By utilizing standard performance metrics, the study demonstrates that the NHBi-RNN framework outperforms existing algorithms in multi-class sentiment analysis, offering a more detailed and accurate interpretation of consumer opinions.

The study [21] demonstrates the importance of analyzing social media data for mental health research by using Twitter data and four BERT models from Hugging Face. The research achieved over a 6% improvement in predictive metrics for depression symptoms compared

to previous studies, evaluated through tenfold cross-validation. By examining tweets and bio information, the study highlights the potential of leveraging social media for early detection and intervention in mental health.

The research [22] utilizes the Sentiment 140 dataset to analyze the performance of a proposed gated attention recurrent network (GARN) architecture for sentiment classification. The dataset is preprocessed to clean and filter the data, followed by feature extraction using a log term frequency-based modified inverse class frequency (LTF-MICF) model. A hybrid mutation-based white shark optimizer (HMWSO) is employed for feature selection. The GARN architecture combines recurrent neural networks (RNN) and attention mechanisms to classify sentiment into positive, negative, and neutral categories. The study also compares the GARN with other deep learning models, such as Bi-GRU, Bi-LSTM, RNN, and CNN, highlighting its effectiveness for Twitter sentiment analysis.

The study [23] employs an ensemble-based deep learning method for developing a recommendation model that simultaneously analyzes ratings and reviews. The process involves identifying features from product reviews and ratings, conducting sentiment analysis to obtain polarity and subjectivity, and scaling inputs to ensure consistency. The recommendation system utilizes deep neural networks with multiple hidden layers and varying nodes, leveraging backpropagation for rapid learning. The model's performance is assessed based on architectural variations of hidden layers using precision, loss, and execution time.

This research [24] introduces a light deep learning (LightDL)-based recommender system using Twitter reviews. The process involves collecting and cleaning Twitter data, then feeding it into the LightDL model, which extracts features such as hashtags, unigrams, and multi-grams. The model categorizes data into positive, negative, and neutral sentiments based on these features. The performance of the LightDL model is evaluated through extensive experiments in MATLAB, assessing metrics such as accuracy, precision, recall, and f-measure.

Shankar, A [25] proposes an intelligent recommendation system utilizing ensemble learning to address common issues in traditional recommendation systems, such as flooding users with irrelevant suggestions and handling new products unpredictably. By incorporating various user activities like browsing history, wish lists, reviews, and ratings in addition to purchase history, the system significantly improves precision and recall. This approach offers a more comprehensive solution compared to conventional techniques, enhancing the overall effectiveness of recommendation systems in e-commerce.

The study [26] proposes a novel recommendation system (RS) that integrates collaborative filtering with sentiment analysis (SA) to enhance the accuracy and personalization of product recommendations in e-commerce. Traditional RS models often rely solely on numerical ratings, which may overlook the subtleties of user preferences. To address this, the study employs a fine-tuned BERT model for sentiment classification and uses its insights to improve the RS's product selection process. The approach demonstrates significant improvements in recommendation accuracy and reliability, outperforming state-of-the-art models in the e-commerce domain.

Mehdi Elahi [27] introduced a hybrid recommendation system (RS) that enhances recommendations by integrating sentiment analysis (SA) of user reviews. The study found that sentiment, which may not always align with ratings, provides an alternative signal for user preferences. The proposed system, tested on Amazon Digital Music and Video Games datasets, outperformed baselines by using a BERT-based NLP pipeline for embedding extraction, followed by PCA for dimensionality reduction, alongside item-based collaborative filtering, ALS, YoutubeRanker, and DeepFM. Similarly, Avishek Garain's hotel RS [28] combines sentiment analysis with aspect-based categorization of Tripadvisor reviews, employing a BERT model, fuzzy logic, and cosine similarity. The system categorizes reviews and gen-

erates recommendations based on user preferences and locations, achieving high accuracy and compact clustering with K-RMS.

Elham Asani [29] developed a context-aware recommendation system that extracts food preferences from restaurant reviews to offer personalized restaurant suggestions. Utilizing natural language processing, the system identifies food items and clusters them using semantic similarity and the Wu–Palmer method, achieving higher precision. Sentiment analysis further classifies user opinions as positive or negative. The system considers user preferences along with factors like location and time to recommend suitable restaurants, demonstrating superior performance in evaluations using 2018 TripAdvisor data. In a related study [30], the paper enhances recommendation systems by integrating sentiment analysis with the k-nearest neighbors collaborative filtering algorithm. It introduces a sentiment rating approach that replaces numerical ratings with sentiment scores and a new sentiment-based user similarity measure called ARP. Evaluated on Yelp and Datafiniti datasets, ARP outperforms traditional similarity measures such as Pearson correlation coefficient and cosine, showing promise in improving recommendation systems.

This research [31] introduces a novel recommendation method that integrates sentiment analysis (SA) with genre-based similarity in collaborative filtering. By employing BERT for genre preprocessing and deep learning models for SA, the method significantly improves performance on movie datasets. This user-based collaborative filtering approach aims to enhance reliability beyond traditional methods like SVD and NMF through the use of SA and genre embedding.

Similarly, the study [32] presents a multilingual recommendation system (RS) with SA designed to help Algerian users make informed decisions based on online reviews. Utilizing semi-supervised support vector machines (SVM) for sentiment scoring, the system effectively analyzes Algerian reviews and provides accurate recommendations, demonstrating its efficiency across languages with user-based collaborative filtering and Spearman similarity. This article [33] presents a hybrid movie recommendation system (RS) that combines collaborative filtering, content-based filtering, and sentiment analysis (SA) of tweets from microblogging sites. Using the MovieTweatings database and focusing on post-2014 movies, the system applies VADER for sentiment analysis of user tweets to enhance recommendations. The approach integrates content-based features with collaborative social filtering, using a weight matrix to compute item similarities based on metadata and user preferences, resulting in robust movie recommendations.

In a related study [34], a framework is introduced that merges SA with a hybrid RS to recommend unreleased movies. The framework analyzes user comments on movie trailers from Netflix's YouTube channel to predict ratings and uses data from The Movie Database (TMDb) to generate a list of preferred upcoming movies for users. By combining sentiment-based ratings with user preferences, the system provides personalized recommendations, utilizing VADER and TextBlob for sentiment analysis of social media comments.

This study [35] enhances recommendation system accuracy by combining sentiment analysis using TextBlob on Twitter data with collaborative filtering. This approach addresses sparsity and cold start problems by integrating user ratings with sentiment insights from tweets, thereby improving the overall recommendation process.

In our previous study [36], we integrated sentiment analysis into recommendation systems using the Yelp dataset, divided into two domains: `yelp_restaurant` and `yelp_hotel`. We employed BERT for sentiment analysis and DeepMF for collaborative filtering, with K-means and AgglomerativeClustering for content-based filtering in the respective domains. These approaches were combined using feature combination to enhance the overall recommendation performance. One key challenge was hyperparameter tuning and the computational

resources required for training BERT. Future work suggested exploring alternative models and improving BERT's tuning.

In contrast, the current study significantly advances our previous research by providing a more comprehensive analysis of the Yelp dataset without dividing it into specific domains, allowing for a more global treatment of the data. Additionally, we enhanced the sentiment analysis process by comparing lexicon-based models with BERT, focusing on hyperparameter optimization to minimize computational demands, thus overcoming one of the major challenges identified in our earlier work. For content-based filtering, we dealt with high-dimensional data by using principal component analysis (PCA) to reduce its complexity. This helped simplify the data while keeping important information, validated through Trustworthiness metrics. We used Word2Vec embeddings for text features. For content-based filtering, we used K-means clustering, while cosine similarity was used for collaborative filtering, with sentiment scores added. The model was tested using t-tests and fivefold cross-validation, which improved the accuracy of the recommendations.

Notably, we proposed a hybrid recommendation system combining collaborative filtering (CF), content-based filtering (CB), and sentiment analysis (SA) using NMF and decision tree regressor. This approach outperformed several techniques, including weighted hybrid methods, yielding the best results with NMF combined with decision tree regressor.

Overall, these improvements not only address the limitations of our earlier study but also lead to more effective recommendations, demonstrating the potential of our current methodology. Tables 1 and 2 present a comparative overview of recommendation systems incorporating sentiment analysis (SA). It details various approaches, including collaborative filtering, content-based models, and sentiment analysis techniques, across different studies. Key methods include item-based filtering, BERT-based preprocessing, and multilingual recommendations. Each approach is evaluated for its strengths and limitations, such as computational demands, domain applicability, and sensitivity to sentiment analysis inaccuracies. The proposed work aims to integrate multiple techniques to enhance recommendation accuracy while addressing challenges in hyperparameter tuning and resource requirements.

3 Proposed system

This section presents the proposed recommendation system, which integrates collaborative filtering, content-based filtering, and sentiment analysis techniques to create a hybrid recommendation system. In order to achieve this, clustering was utilized for content-based filtering, while memory-based similarity measure was employed for collaborative filtering. The primary objective is to enhance the reliability of user recommendations by incorporating sentiment analysis of user reviews or comments alongside traditional recommendation methods. Figure 1 illustrates the architecture of this system.

3.1 Dataset description

The Yelp dataset [37], released in 2017, offers a rich collection of information about businesses and user reviews. It includes attributes such as business name, location (city, latitude, longitude, full address, and state), business type, operational status, review count, star rating, and categories. Reviewer-related attributes cover reviewer name, type, average star rating, and review count. The dataset also provides review-specific details including review ID, date, text, star rating, and metrics for usefulness, coolness, and funniness. The dataset contains

Table 1 Comparative overview of recommendation systems incorporating sentiment analysis

Paper	Collaborative filtering model	Content-based model	Sentiment analysis model	Dataset	Proposed approach	Limitations
[27]	Item-based Collaborative Filtering, ALS, YoutubeRanker, DeepFM	N/A	BERT-base-uncased for last layer representations	Amazon Digital Music and Video Games datasets	Hybrid RS with SA	The system lacks content-based filtering, limiting its personalization capabilities for new users or items
[29]	Item-to-item Collaborative Filtering with Cosine Similarity	Semantic similarity, Wu–Palmer clustering	SentiWordNet-based	TripAdvisor data	Hybrid RS extracting food preferences from user comments	Limited applicability to domains other than restaurants, such as movies or books
[30]	k-Nearest Neighbors Collaborative Filtering	N/A	SentiWordNet-based	Yelp and Datafiniti datasets	Sentiment-based user similarity measure (ARP)	Sensitivity to inaccuracies in SA, e.g., SentiWordNet
[31]	User-based Collaborative Filtering with BERT-based Genre Preprocessing	N/A	BERT-based feature vectors; LSTM+CNN models for SA	movie datasets	Hybrid RS with sentiment analysis and genre-based similarity	High computational demands of deep learning models

Table 2 Comparative overview of recommendation systems incorporating sentiment analysis

Paper	Collaborative filtering model	Content-based model	Sentiment analysis model	Dataset	Proposed approach	Limitations
[26]	User-based CF, Item-based CF	N/A	BERT	Amazon Musical Instruments reviews dataset	Hybrid RS with SA	Not using content-based filtering may reduce the system's ability to capture specific item features, potentially affecting recommendation precision
[32]	User-based Collaborative Filtering with Spearman Similarity	N/A	Semi-supervised support vector machines (SVM)	Restaurant Tijuana Restaurant dataset, french dataset collected from ldle.com, Arabic and dialect dataset collected from dz.jumia.com	Multilingual recommender with SA for Algerian users	Challenges with semi-supervised SVM and generalization
Our Proposed Work	Cosine Similarity with SA	Clustering using K-means, Word2Vec, StandardScaler, PCA	BERT for SA (AdamW, LR 1e-5)	Yelp Dataset	Hybrid approach with clustering, cosine similarity and BERT using NMF, and DecisionTreeRegressor	Hyperparameter tuning challenges; high computational resources for BERT training

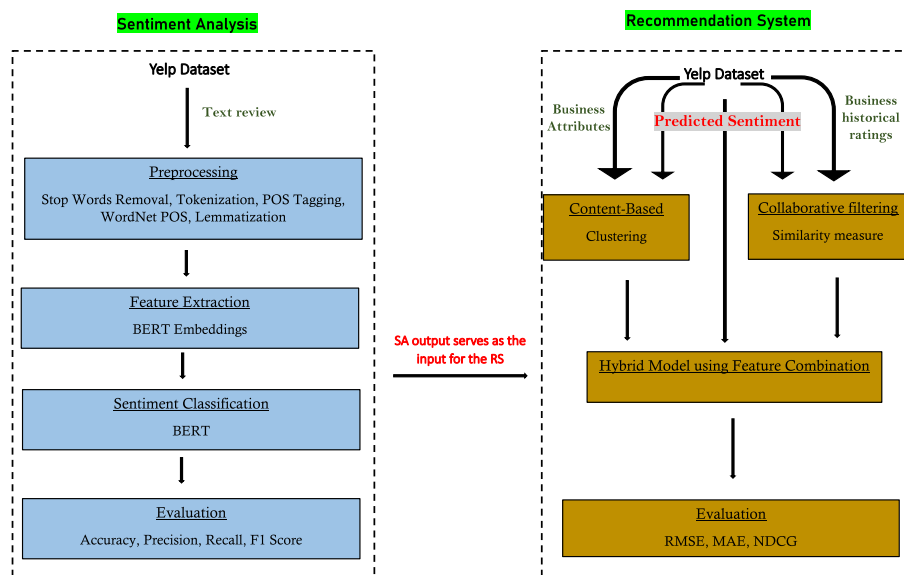


Fig. 1 A hybrid recommendation framework enhanced with sentiment analysis

Table 3 Details of the Yelp dataset

Metric	Value
Unique Businesses	2783
Unique Users	3676
Total Reviews	5000
Total Funny Votes	3436
Total Cool Votes	4350

5,000 reviews across 2,783 unique businesses and 3,676 unique users, as summarized in Table 3. Key features extracted from the dataset are illustrated in Table 4, which includes example data for various attributes such as business ID, latitude, longitude, review text, and reviewer statistics.

3.2 Preprocessing of review text data

Effective sentiment analysis necessitates thorough preprocessing to transform raw text into a structured format suitable for analysis. The preprocessing steps for the Yelp review data include:

1. **Stop Words Removal:** Common words such as “the,” “is,” and “and,” which do not add significant meaning to the analysis, were removed to reduce noise in the data.
2. **Tokenization:** The text was divided into individual tokens or words, facilitating detailed analysis by breaking down the text into manageable units.
3. **Part-of-Speech Tagging:** Each token was assigned a grammatical tag (e.g., noun, verb, adjective) to better understand the text’s grammatical structure and context, which aids in more nuanced sentiment classification.

Table 4 Example of yelp dataset features extracted

Features	Value
business_blank	False
business_categories	Greek; Restaurants
business_city	Scottsdale
business_full_address	1336 N Scottsdale Rd Scottsdale, AZ 85257
business_id	-5jkZ3-nUPZxUvtcbr8Uw
business_latitude	33.463373
business_longitude	-111.926908
business_name	George's Gyros Greek Grill
business_open	True
business_review_count	11
business_stars	4.5
business_state	AZ
business_type	business
cool	0
date	2012-05-22
funny	0
review_id	rAS26HNMyqj_d6PBf4vuHw
reviewer_average_stars	3.83
reviewer_blank	False
reviewer_cool	1
reviewer_funny	1
reviewer_name	Natalie
reviewer_review_count	6
reviewer_type	user
reviewer_useful	9
stars	4
text	Good gyros, clean and friendly staff
type	review
useful	1
user_id	S_NH8JMiL0v9mrNBcClp0Q

4. WordNet Part-of-Speech Information: Part-of-speech tagging was aligned with WordNet's lexical database, which enhances the system's understanding of word relationships and meanings, contributing to more accurate sentiment analysis.
5. Lemmatization: Words were reduced to their base or root forms, standardizing the text and minimizing the number of unique words, which helps in improving the consistency and accuracy of sentiment classification.

These preprocessing steps ensure that the text data is clean, well structured, and optimized for effective sentiment classification, addressing potential issues like noise and variability in the data.

Figure 2 illustrates a review before and after preprocessing, highlighting the transformation of raw text into a more streamlined format.

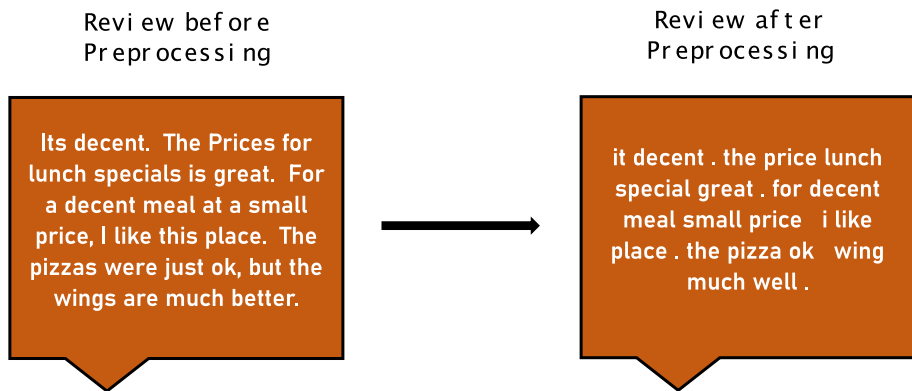


Fig. 2 Effect of preprocessing on review text

3.3 Sentiment analysis

In our study, we utilized the BERT (Bidirectional Encoder Representations from Transformers) model to conduct sentiment analysis, categorizing sentiments into two primary classes: “Negative” and “Positive.” BERT’s architecture is particularly suited for this task due to its ability to capture contextual information from text, making it effective in understanding nuances in sentiment expression. To optimize the model’s performance, we selected several key hyperparameters: a learning rate of 1×10^{-5} , which ensures gradual and stable updates during training; a batch size of 16, allowing the model to process a manageable number of samples at once; and a weight decay of 0.01 to prevent overfitting by penalizing overly complex models. We conducted training for a single epoch to gauge initial performance, adjusting parameters based on validation results.

For our implementation, we employed a tokenizer specifically designed for BERT, which is crucial for preparing text data. This tokenizer transforms raw text into subword units, enabling the model to effectively process and understand the input. It handles essential tasks such as lowercasing and adding special tokens that are necessary for the BERT model to function correctly. By utilizing a pre-trained tokenizer, we can ensure that the input is formatted consistently with the training data that BERT has previously encountered.

In addition, we initialized a pre-trained BERT model configured for sequence classification tasks, tailored to output two labels for our binary sentiment classification. Leveraging a pre-trained version of BERT allows us to take advantage of the extensive knowledge it has acquired from training on a large corpus of text, thereby enhancing its ability to grasp context and sentiment nuances.

Furthermore, we prepared our dataset by splitting it into training and testing sets, with 30% allocated for testing. This division is crucial for evaluating the model’s performance on unseen data and ensuring that our results are generalizable. By carefully configuring these parameters and utilizing BERT’s powerful architecture, including its tokenizer and model capabilities, we aimed to achieve an accurate classification of sentiments, providing valuable insights into the emotional landscape of the data (Fig. 3).

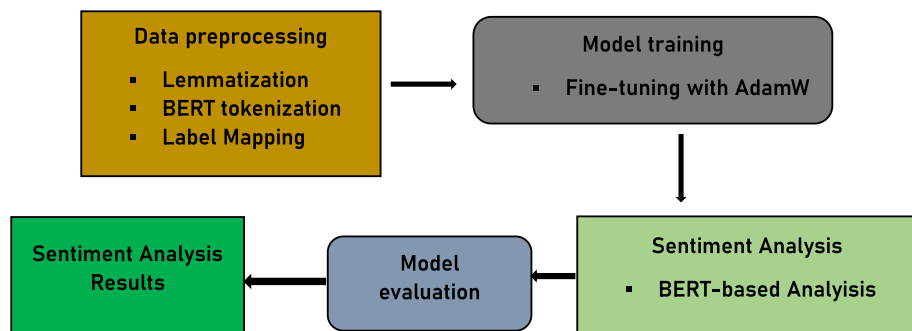


Fig. 3 BERT-based sentiment analysis architecture

3.4 Hybrid recommendation system

Our proposed hybrid recommendation system for Yelp combines collaborative filtering and content-based filtering techniques. Collaborative filtering takes into account user preferences and sentiments predicted from ratings and reviews, while content-based filtering clusters businesses based on attributes such as business category, name, address, city, and review count. By leveraging both methods, our system provides personalized recommendations tailored to user preferences, helping them discover new business that align with their tastes.

3.4.1 Collaborative filtering

- **Cosine similarity** is a widely utilized metric for quantifying the similarity between two vectors within a multi-dimensional space. It calculates the cosine of the angle between the vectors, providing a measure of their alignment in terms of direction. In recommendation systems, cosine similarity is commonly employed to determine the similarity between user preferences or item features [38]. The cosine similarity score ranges from -1 to 1, where a value of 1 signifies perfect similarity, 0 indicates no similarity, and -1 suggests complete dissimilarity. However, in our work, which employs nonnegative ratings, cosine similarity is interpreted within the range of 0 to 1, where higher values indicate a stronger resemblance between vectors, reflecting a greater likelihood of shared characteristics or preferences.

In our approach, cosine similarity was applied to the user–item interaction matrix to measure the similarity between users (or items). Specifically, for each pair of users (or items), we computed the cosine similarity score based on their interaction vectors. These vectors represent the users’ preferences or the items’ features across all available items or users, respectively. The similarity was computed using the following equation:

$$\text{Similarity}(\mathbf{A}, \mathbf{B}) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \cdot \|\mathbf{B}\|} \quad (1)$$

Here, $\mathbf{A}$ and $\mathbf{B}$ represent the interaction vectors for two users or items. The dot product $\mathbf{A} \cdot \mathbf{B}$ calculates the magnitude of their overlap, while $\|\mathbf{A}\|$ and $\|\mathbf{B}\|$ are the Euclidean norms that normalize the vectors.

- **Integration of Sentiment Analysis:** In addition to traditional collaborative filtering, our approach incorporates sentiment analysis predictions derived from the BERT model, evaluated using fivefold cross-validation to ensure robust performance and generalization.

– *Sentiment Predictions using BERT:*

1. *BERT for Sentiment Analysis:* The BERT model was utilized to analyze textual reviews and predict whether the sentiment behind each review was positive or negative. These predictions were crucial because they provided insights into the subjective feedback expressed in user reviews, which is valuable beyond the explicit star ratings.
2. *Normalization of Sentiment Scores:* Since BERT outputs binary sentiment labels (positive or negative), we converted these sentiment predictions into a continuous scale ranging from 1 to 5 to align them with the star ratings (explicit numerical ratings). This normalization was achieved using a MinMaxScaler, ensuring that sentiment scores were comparable to traditional rating values in the collaborative filtering system.

– *Collaborative Filtering Model with Sentiment Integration*

1. *Traditional Collaborative Filtering:* Collaborative filtering uses user–item interaction data to generate recommendations by computing similarities between users or items. We use a user-based cosine similarity model to measure how similar users are based on their ratings, where higher similarity scores indicate more aligned preferences for improved recommendations.
2. *Sentiment as an Additional Feature:* The BERT sentiment predictions, which represent the emotional content of user reviews, are integrated as additional features into the collaborative filtering process. Specifically, these sentiment scores are normalized and added to the user–item interaction matrix (as part of the raw ratings). The collaborative filtering model will utilize both user ratings and the normalized sentiment scores. This integration allows the model to adjust the way recommendations are made by considering the sentiment expressed in user reviews alongside numerical ratings. By incorporating both explicit numerical ratings and implicit sentiment data, the model can predict user preferences more effectively, leading to more personalized and accurate recommendations.

3.4.2 Content-based

In our content-based recommendation system, we employ clustering techniques to group items based on their textual and numerical attributes, including sentiment scores predicted from the text. The process involves the following steps:

1. *Textual attribute representation:* We use Word2Vec (Skip-gram) to generate vector representations for the textual attributes of the items. Word2Vec is selected because it effectively captures semantic relationships between words, allowing us to generate meaningful vector embeddings for textual data, which are essential for capturing nuanced information in item descriptions.
2. *Numerical attribute normalization:* Numerical attributes, such as business review count and predicted sentiment, are standardized using StandardScaler. Standardization is essential to ensure that all features contribute equally to the clustering process, preventing attributes with larger scales from dominating the analysis.
3. *Attribute combination and dimensionality reduction:* The combined attribute vectors (textual and numerical) are then subjected to dimensionality reduction through principal component analysis (PCA). PCA is used to reduce the feature space while preserving the most important information, which helps in simplifying the clustering process and improving computational efficiency.

4. *Clustering*: The transformed attribute vectors are fed into the K-means clustering algorithm to group similar items together. Through experimentation, we found K-means to be the most effective algorithm compared to MiniBatchKMeans and KMedoids, as evaluated by the silhouette score. K-means provided the best balance between computational efficiency and clustering quality, making it ideal for our large dataset.

By incorporating this clustering-based approach, our system can offer personalized recommendations by considering both textual and numerical item attributes. This enables a more comprehensive understanding of item characteristics and user preferences, leading to more relevant and accurate recommendations.

The grouping of items based on their attributes allows for the identification of similarities and patterns, enhancing the system's ability to understand user preferences and make accurate recommendations. Combining textual and numerical attributes further enriches the recommendation process, providing a holistic view of item characteristics.

Overall, the integration of clustering techniques in our content-based recommendation system enhances the effectiveness and personalization of the recommendations, improving the overall user experience (Fig. 4).

3.4.3 Hybrid recommendation system: NMF and DecisionTreeRegressor fusion

In recommender systems, a common approach to improving accuracy and personalization is to combine multiple techniques, creating a hybrid model. There are various types of hybrid models, such as the weighted hybrid, which linearly combines the outputs of different recommendation techniques, the switching hybrid, which selects the most appropriate technique for each recommendation, and the feature combination hybrid, which integrates different features or algorithms into a single model.

The feature combination hybrid approach is particularly effective as it allows the recommender system to leverage the complementary strengths of different techniques. By combining multiple features or algorithms, the hybrid model can capture more comprehensive information about user preferences and item characteristics, leading to more accurate and personalized recommendations.

In our work, we propose a feature combination hybrid recommendation system that combines nonnegative matrix factorization (NMF) and DecisionTreeRegressor. NMF is a technique used in data analysis and machine learning to extract meaningful patterns from nonnegative data. It decomposes a matrix into two lower-rank matrices, capturing positive relationships and dependencies [39]. NMF has been effective in enhancing tasks like collaborative filtering, content-based filtering, and sentiment analysis.

We utilize NMF to extract latent factors from the data, capturing underlying patterns and relationships between user preferences and item characteristics. The NMF-transformed features are then combined with other relevant features, such as collaborative filtering cosine predictions, clustering results, and BERT predictions, and fed into the DecisionTreeRegressor algorithm. By leveraging the strengths of both NMF and DecisionTreeRegressor, we aim to create a more accurate and personalized recommendation model that effectively captures complex relationships and improves the overall performance of our recommendation system. This feature combination hybrid approach allows us to take advantage of the complementary strengths of NMF and DecisionTreeRegressor, leading to a more comprehensive and effective recommendation system compared to using either technique alone.

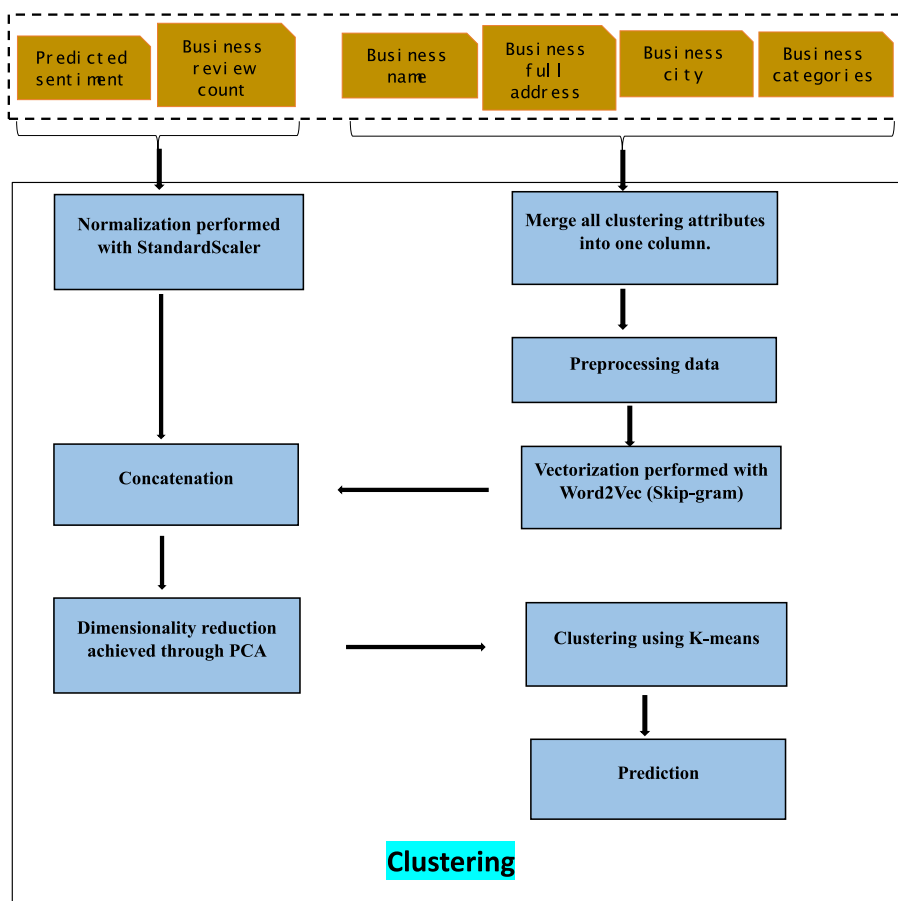


Fig. 4 Clustering pipeline for item analysis

In architecture 5, we have clarified that the outputs from collaborative filtering (CF), content-based filtering (CB), and sentiment analysis (SA) serve as inputs for the hybrid recommendation system (HRS), specifically in the feature combination hybrid model.

1. *Feature combination*: The outputs from collaborative filtering, content-based filtering, and sentiment analysis are combined into a single feature matrix. This step ensures that the model leverages all available information from different recommendation techniques.
2. *Feature transformation (NMF)*: Nonnegative matrix factorization (NMF) is applied to the combined features, transforming them into a lower-dimensional space. This step helps in uncovering latent factors that represent underlying patterns in user preferences.
3. *DecisionTreeRegressor*: Finally, the transformed features, along with the original features, are fed into a DecisionTreeRegressor. This model learns from the enriched feature set to make the final predictions, optimizing the recommendation accuracy.

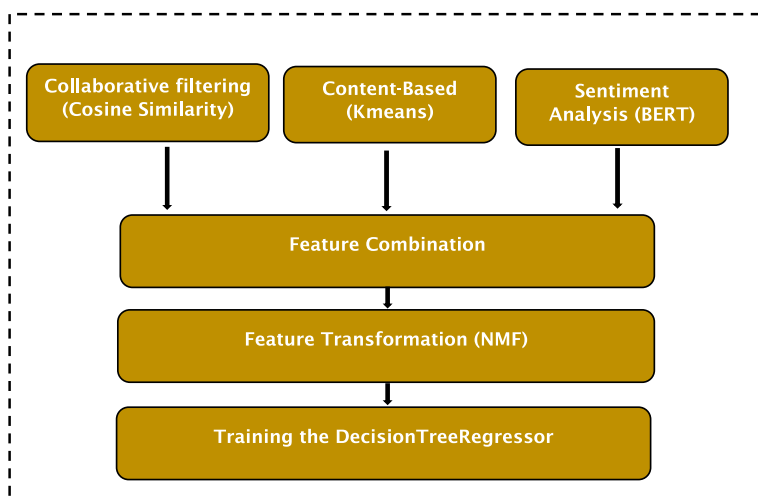


Fig. 5 Hybrid recommender system using NMF and DecisionTreeRegressor

Algorithm 1 Hybrid Recommendation System with Sentiment Analysis Integration

- 1: **Input:** Collaborative Filtering predictions ($CF_Predictions$), Content-Based Filtering predictions ($CB_Predictions$), Sentiment Analysis predictions using BERT ($SA_Predictions$), Target variable: User ratings ($Stars$)
- 2: **Output:** Final recommendations, Evaluation metric (RMSE)
- 3: **Step 1: Preprocessing**
- 4: Handle missing values in $CF_Predictions$, $CB_Predictions$, and $SA_Predictions$:
- 5: Replace missing values with the mean of the respective column.
- 6: Normalize $CF_Predictions$, $CB_Predictions$, and $SA_Predictions$:
- 7: Scale the values to the range $[1, 5]$ using MinMaxScaler.
- 8: **Step 2: Feature Combination**
- 9: Combine $CF_Predictions$ using cosine similarity,
- 10: $CB_Predictions$ generated from K-Means clustering, and
- 11: $SA_Predictions$ from BERT into a single feature matrix ($Feature_Matrix$).
- 12: **Step 3: Feature Transformation**
- 13: Apply Non-Negative Matrix Factorization (NMF) on $Feature_Matrix$:
- 14: Transform features into a lower-dimensional space to extract latent factors ($NMF_Features$).
- 15: **Step 4: Feature Integration**
- 16: Concatenate the original features ($Feature_Matrix$) with $NMF_Features$ to form a hybrid feature set ($Hybrid_Features$).
- 17: **Step 5: Model Training**
- 18: Initialize DecisionTreeRegressor.
- 19: Train the DecisionTreeRegressor on $Hybrid_Features$ with $Stars$ as the target variable.
- 20: **Step 6: Prediction**
- 21: Use the trained DecisionTreeRegressor to predict user ratings ($Predictions$).
- 22: **Step 7: Evaluation**
- 23: Compute the root mean square error (RMSE) between the actual user ratings ($Stars$) and $Predictions$.
- return** $Predictions$, $RMSE$

4 Results and discussion

4.1 Metrics

The metrics employed to assess the accuracy and reliability of rating predictions include:

1. Accuracy is the ratio of correctly predicted instances (both true positives and true negatives) to the total number of instances.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (2)$$

2. Precision is the ratio of correctly predicted positive instances (true positives) to the total predicted positive instances.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (3)$$

3. Recall, also known as sensitivity or true positive rate, is the ratio of correctly predicted positive instances to all instances that actually belong to the positive class.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (4)$$

4. The F1 score is the harmonic mean of precision and recall.

$$\text{F1 score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

5. The silhouette score measures how similar an object is to its own cluster compared to other clusters. It is defined for each point as:

$$\text{Silhouette Score} = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (6)$$

where:

- $a(i)$ is the average distance between the point i and all other points in the same cluster.
 - $b(i)$ is the minimum average distance between the point i and all points in any other cluster.
6. RMSE short for root mean square error, quantifies the average discrepancy between predicted and actual values, serving as a metric to evaluate a model's performance in regression analysis. It is calculated by taking the square root of the mean of squared prediction errors. A lower RMSE suggests higher accuracy and a better fit between predicted and actual values [40].

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (7)$$

In this equation, n is the total number of observations, y_i represents the actual or observed values, and $\hat{y}_i$ represents the predicted values.

7. MAE, or mean absolute error, measures the average absolute difference between predicted and actual values, providing an assessment of a model's accuracy, with lower values

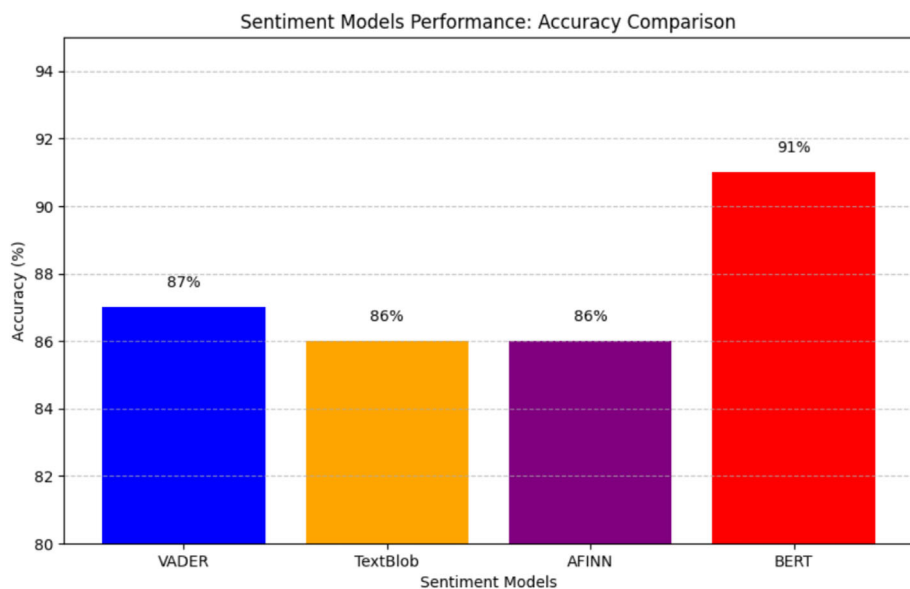


Fig. 6 Sentiment model performance: accuracy comparison

indicating better performance and a closer match between predictions and observed data [41].

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (8)$$

In this equation, n is the total number of observations, y_i represents the actual or observed values, and $\hat{y}_i$ represents the predicted values.

4.2 Sentiment analysis results

In the results of the sentiment analysis (SA) shown in Fig. 6, we evaluated the performance of multiple algorithms. The VADER algorithm achieved an accuracy of 87%, while both TextBlob and AFINN achieved an accuracy of 86%. Additionally, BERT, a state-of-the-art language model, outperformed the other algorithms with an impressive accuracy of 91%.

The sentiment analysis results, illustrated in Fig. 7, demonstrate that BERT outperformed the other methods, achieving a precision of 91%, while Vader, TextBlob, and AFINN each reached 88% precision. This highlights BERT's superior ability to understand contextual nuances, enabling more accurate sentiment classification compared to the traditional rule-based approaches of Vader and AFINN, and the simpler machine learning method of TextBlob. The consistent precision among Vader, TextBlob, and AFINN suggests that while these methods are effective, they may struggle to capture more complex sentiment patterns that BERT successfully identifies.

The recall results, depicted in Fig. 8, reveal that BERT achieved the highest recall at 98%, surpassing Vader (87%), TextBlob (96%), and AFINN (97%). This indicates BERT's superior capability in identifying true positive sentiment cases, making it more effective in capturing the full spectrum of sentiment within the text. While TextBlob and AFINN also demonstrated

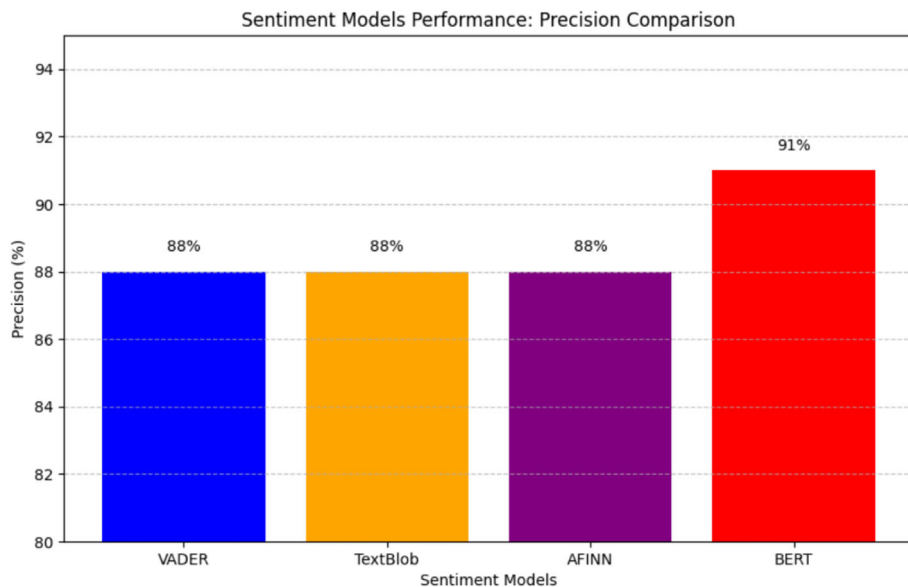


Fig. 7 Sentiment models performance: precision comparison

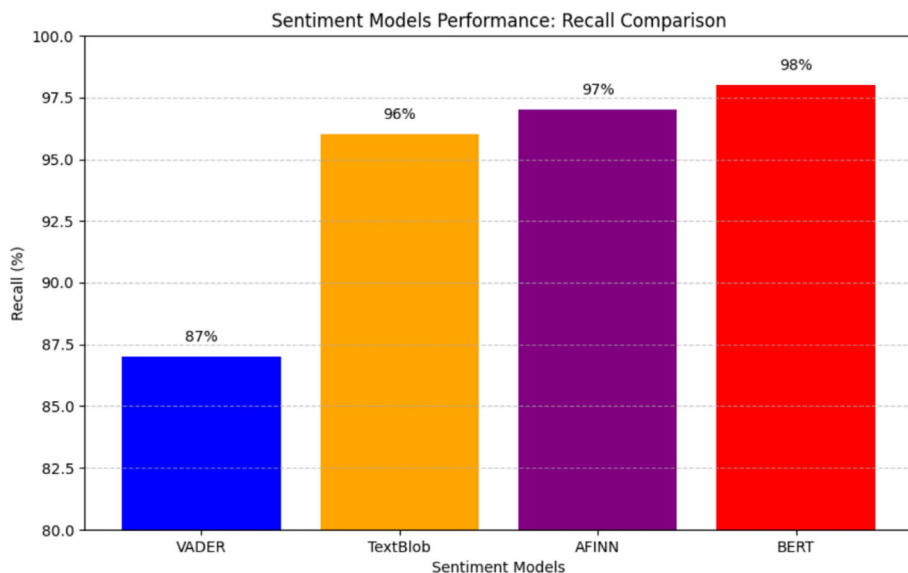


Fig. 8 Sentiment models performance: recall comparison

strong recall performance, the lower recall of Vader suggests it may miss certain sentiment cues that the others successfully detect. BERT's high recall reflects its advanced contextual understanding, which is essential for comprehensive sentiment analysis.

The F1 score results, shown in Fig. 9, indicate that BERT achieved the highest F1 score at 94%, slightly ahead of Vader, TextBlob, and Afinn, which each scored 92%. This suggests that BERT offers the best balance between precision and recall, making it the most reliable model

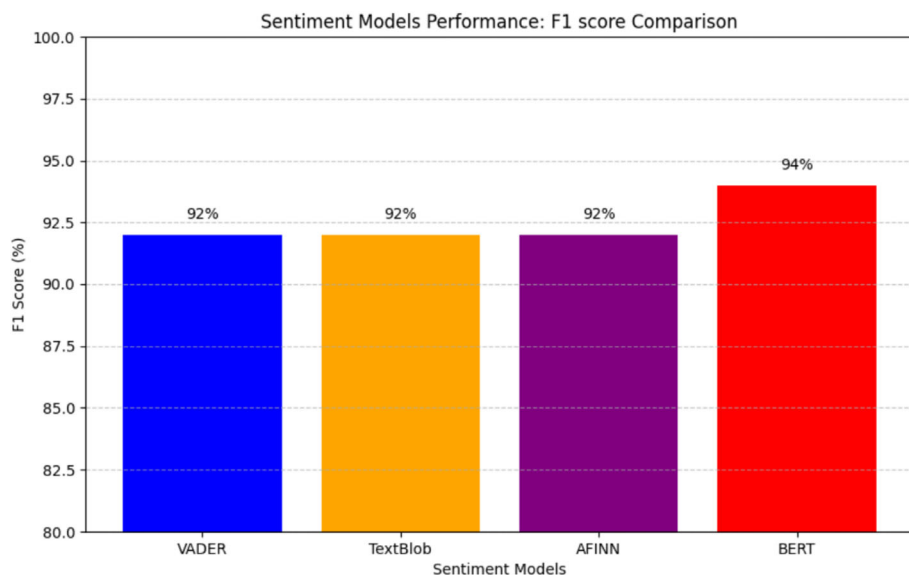


Fig. 9 Sentiment models performance: F1 score comparison

for overall sentiment classification. While Vader, TextBlob, and AFINN also performed well, BERT's slight edge in F1 score highlights its enhanced ability to accurately and consistently identify sentiments, making it the most effective model for this task.

Table 5 presents a detailed comparison of the performance metrics for the BERT model in sentiment analysis, focusing on three distinct hyperparameter configurations. Each configuration varies key parameters such as learning rate, batch size, weight decay, and the optimizer used. The first configuration, with a learning rate of 1×10^{-5} , a batch size of 8, and a weight decay of 0.01, achieves an accuracy of 89.85%, precision of 90.95%, recall of 97.61%, and an F1 score of 94.16%. In contrast, the second configuration, which adopts a learning rate of 5×10^{-5} and a batch size of 16 with a weight decay of 0.1, yields slightly improved results, with an accuracy of 90% and precision of 93%. Notably, the third configuration with a learning rate of 1×10^{-5} and a batch size of 16 demonstrates superior performance across all metrics, achieving an accuracy of 91% and a remarkable recall of 98%. This indicates that fine-tuning hyperparameters is crucial in optimizing model performance, particularly in enhancing recall, which is vital for sentiment analysis tasks where capturing true positive sentiments is essential. Overall, the results underscore the importance of careful hyperparameter selection in machine learning models to achieve optimal performance.

In our pursuit of creating a comprehensive and powerful RS, we seamlessly integrated the BERT predictive model into our content-based (CB) and collaborative filtering (CF) approaches, culminating in a robust hybrid RS that incorporates SA. By combining CB filtering, CF, and SA, we leverage the strengths of each approach to provide personalized and sentiment-aware recommendations. This integration allows us to consider various factors, such as item characteristics, collaborative patterns, and sentiment expressions, resulting in a more holistic understanding of user preferences. By leveraging BERT for sentiment analysis within the hybrid RS, we enhance the accuracy, relevance, and user satisfaction of our recommendations, ultimately delivering an exceptional recommendation experience.

Table 5 Performance comparison of BERT sentiment analysis across different hyperparameter settings

Hyperparameter	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
Learning Rate: 1×10^{-5} , batch size: 8, epochs: 1, weight decay: 0.01, optimizer: AdamW	89.85	90.95	97.61	94
Learning Rate: 5×10^{-5} , batch size: 16, epochs: 1, weight decay: 0.1, optimizer: AdamW	90	93	94	94
Learning Rate: 1×10^{-5} , batch size: 16, epochs: 1, weight decay: 0.01, optimizer: AdamW	91	91	98	94.8

Table 6 Comparison of dimensionality reduction techniques based on trustworthiness

Dimensionality reduction technique	Trustworthiness
PCA	0.617
Isomap	0.565
Kernel PCA	0.576
LLE	0.554
Autoencoder	0.581

4.3 Content-based: clustering result

Table 6 presents the trustworthiness values obtained for different dimensionality reduction techniques (PCA, Isomap, kernel PCA, locally linear embedding (LLE), and autoencoders) applied to the combined feature set of textual and numerical attributes, including sentiment scores.

Dimensionality reduction is a crucial step in this process to address the challenges posed by the high-dimensional data generated from combining textual and numerical features. Textual attributes, represented as Word2Vec embeddings, effectively capture semantic relationships between words, while numerical features, such as business review counts and sentiment scores, are standardized to ensure balanced contributions during analysis. However, this combined feature set creates a complex, high-dimensional space, which can lead to increased computational demands and potential noise. By applying dimensionality reduction techniques, such as PCA and others, we simplify the feature space while retaining the most significant information. This not only enhances computational efficiency but also ensures that the reduced data remains suitable for downstream clustering tasks, as reflected in the trustworthiness values shown in the table.

Trustworthiness is a metric used to evaluate how well a dimensionality reduction technique preserves the local structure of the original high-dimensional data. Specifically, it measures whether the nearest neighbors in the high-dimensional space remain neighbors in the reduced space. A higher trustworthiness score indicates better preservation of the original data's structure [43].

As shown in Table 6, PCA achieved the highest trustworthiness score (0.617) among the evaluated techniques. This result demonstrates that PCA was most effective in preserving the local structure of the data, which is crucial for ensuring meaningful clustering results in the content-based recommendation system.

While techniques like kernel PCA and autoencoder also performed relatively well, their trustworthiness scores were slightly lower than PCAs, indicating a minor loss of structural integrity during dimensionality reduction. Methods like Isomap and LLE exhibited even lower trustworthiness, suggesting that these techniques may not be as suitable for our dataset in this specific context.

Based on these findings, PCA was selected for dimensionality reduction in our proposed method, as it provides a robust balance between computational efficiency and the preservation of the data's intrinsic structure.

In our content-based recommendation system, dimensionality reduction and clustering are closely linked processes that enhance item grouping. Techniques like PCA simplify the high-dimensional feature space while preserving essential data structure, as indicated by trustworthiness scores. This simplification facilitates effective clustering, with the Elbow method helping to identify the optimal number of clusters for algorithms like KMeans,

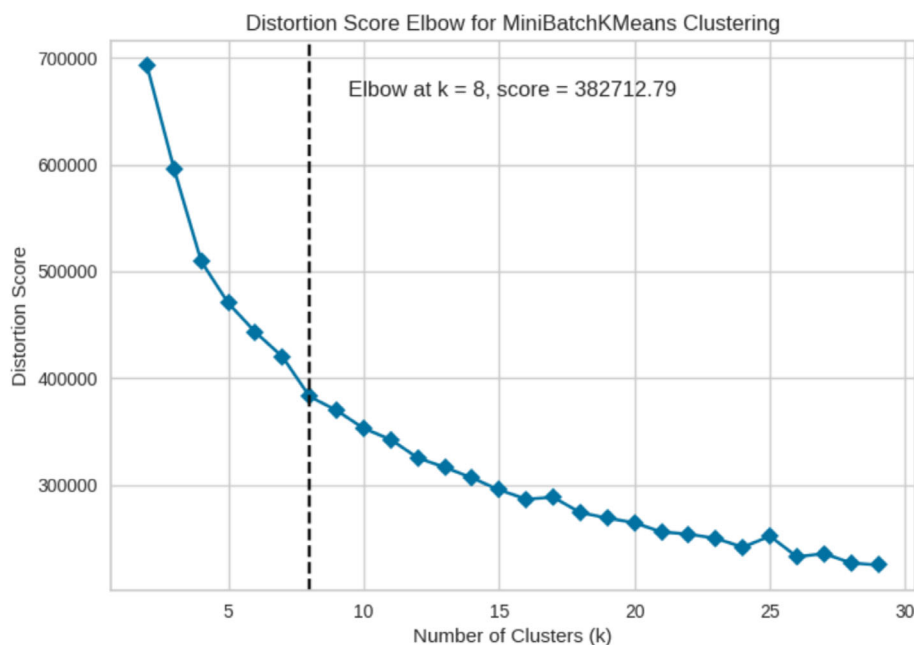


Fig. 10 Elbow method for optimal number of clusters using MiniBatchKMeans

MiniBatchKMeans, and KMedoids (Figs. 10, 11, 12). The corresponding silhouette scores further confirm the quality of the resulting clusters (Table 7), demonstrating the synergy between dimensionality reduction and clustering in improving computational efficiency and the overall effectiveness of our recommendation system.

The results from the elbow method for the three clustering algorithms: MiniBatchKMeans, KMedoids, and KMeans, provide valuable insights into the optimal number of clusters for the given dataset.

For the MiniBatchKMeans algorithm, the elbow is observed at $k = 8$, with a corresponding distortion score of 382712.793. This suggests that increasing the number of clusters beyond 8 yields diminishing returns in terms of reducing the distortion score, indicating that 8 clusters effectively capture the underlying structure in the data.

In the case of KMedoids, the optimal number of clusters is determined to be $k = 10$, with a distortion score of 408966.773. Although this score is higher than that of MiniBatchKMeans, the elbow indicates that using 10 clusters helps to better delineate the data points, especially in scenarios where the medoids can offer robustness against noise and outliers.

Lastly, for KMeans, the elbow is seen at $k = 7$, with a distortion score of 399804.411. This result suggests that a configuration of 7 clusters provides a good balance between model complexity and performance, as it captures the essential patterns in the data without overfitting.

Table 7 presents the silhouette scores for three clustering algorithms: KMeans, MiniBatchKMeans, and KMedoids. The silhouette score is a metric that quantifies how well each data point is clustered, with higher values indicating better defined clusters.

Among the algorithms evaluated, KMeans achieves the highest silhouette score of 0.227, suggesting that it provides the most distinct and well-separated clusters for the dataset. This

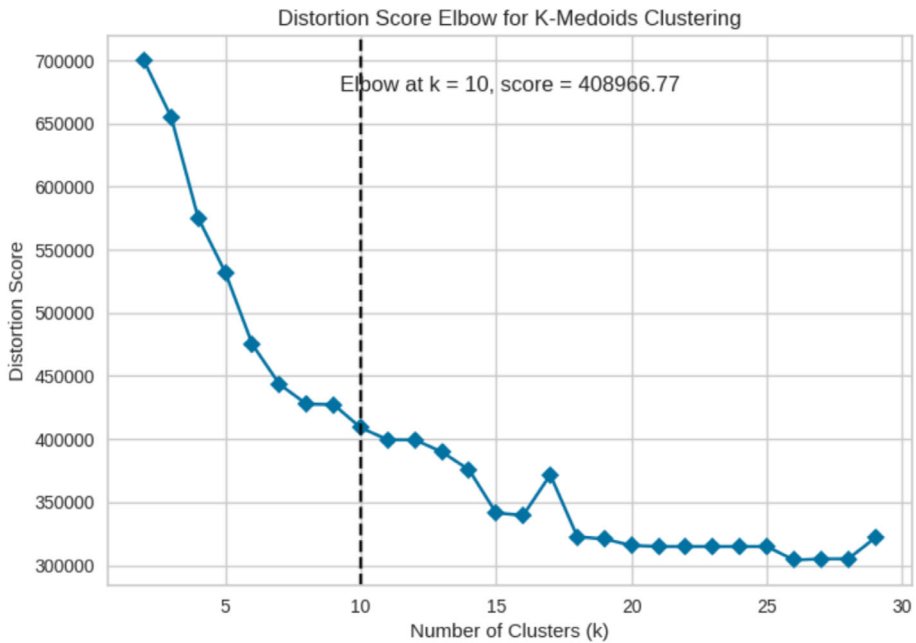


Fig. 11 Elbow method for optimal number of clusters using KMedoids

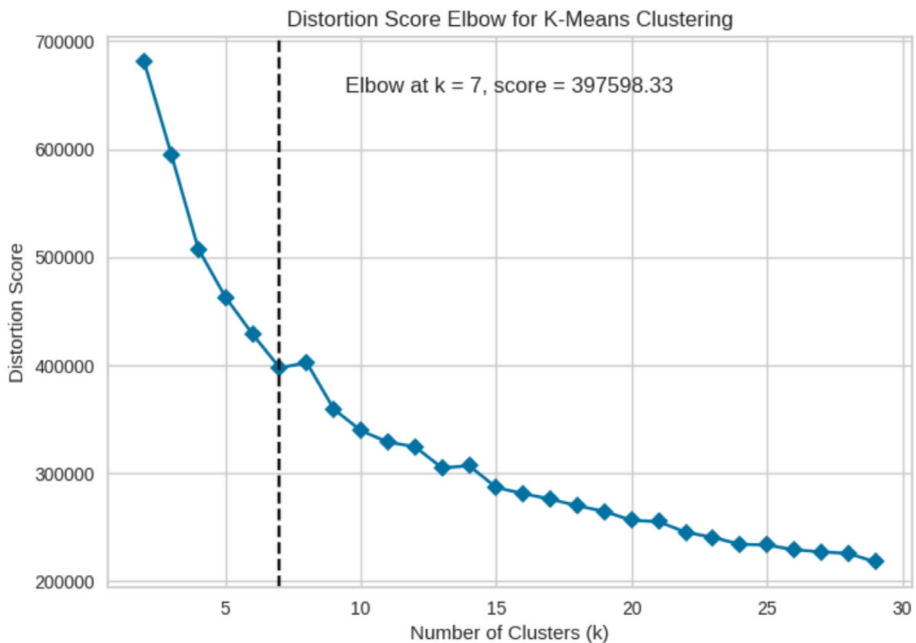


Fig. 12 Elbow method for optimal number of clusters using KMeans

Table 7 Silhouette scores of different clustering algorithms

Clustering algorithm	Silhouette score
KMeans	0.227
MiniBatchKMeans	0.192
KMedoids	0.110

Table 8 Similarity measure evaluation without sentiment analysis (fivefold cross-validation)

Similarity measure	RMSE	MAE
Cosine	1.2703	1.0124
MSD	1.2752	1.0165
Pearson	1.2711	1.0163

indicates that the data points within each cluster are closer to each other compared to points in other clusters, reflecting effective clustering.

In contrast, MiniBatchKMeans has a slightly lower silhouette score of 0.192. While it still indicates some degree of separation, the clusters formed by this algorithm are less cohesive than those created by KMeans. This could be attributed to the stochastic nature of MiniBatchKMeans, which processes data in smaller batches, potentially leading to less precise clustering.

Lastly, KMedoids has the lowest silhouette score of 0.110, indicating that this algorithm struggles to form well-defined clusters within the dataset. The reduced performance may be due to its sensitivity to outliers and noise, which can adversely affect the selection of medoids and the overall clustering outcome.

Overall, the results highlight the varying effectiveness of each clustering algorithm in capturing the underlying structure of the data, with KMeans demonstrating the strongest performance based on the silhouette score metric.

4.4 Collaborative filtering result

Table 8 summarizes the evaluation of various similarity measures used in the recommendation system without incorporating sentiment analysis, assessed through fivefold cross-validation. Among the measures, the cosine similarity achieved the lowest root mean square error (RMSE) of 1.2703 and a mean absolute error (MAE) of 1.0124, indicating its effectiveness in capturing user–item relationships. In comparison, the mean squared difference (MSD) and Pearson correlation yielded slightly higher RMSE values of 1.2752 and 1.2711, respectively, along with corresponding MAE values of 1.0165 and 1.0163. These results suggest that while all similarity measures performed relatively similarly, the cosine similarity was the most effective in minimizing prediction error, highlighting its potential utility in enhancing recommendation accuracy.

The results presented in Table 9 illustrate the evaluation of different similarity measures after incorporating sentiment analysis into the recommendation system, using fivefold cross-validation. Notably, the cosine similarity measure achieved the lowest root mean square error (RMSE) of 1.2478 and the lowest mean absolute error (MAE) of 0.9803, indicating superior performance compared to the other measures. In contrast, the mean squared difference (MSD) and Pearson correlation measures recorded RMSE values of 1.2571 and 1.2668, respectively,

Table 9 Similarity measure evaluation with sentiment analysis (fivefold cross-validation)

Similarity measure	RMSE	MAE
Cosine	1.2478	0.9803
MSD	1.2571	1.0165
Pearson	1.2668	1.0213

Table 10 Performance comparison of hybrid recommendation models

Model	Performance (RMSE)
Weighted Hybrid	1.24
RandomForestRegressor	1.2110
NMF + RandomForestRegressor	1.1970
GradientBoostingRegressor	1.2108
NMF + GradientBoostingRegressor	1.2011
SVR (Support Vector Regressor)	1.2292
NMF + SVR	1.2394
DecisionTreeRegressor	1.2107
NMF + DecisionTreeRegressor	1.1955

along with MAE values of 1.0165 and 1.0213. These results highlight the effectiveness of the cosine similarity measure when sentiment analysis is integrated into the system, suggesting that it may better capture the nuances of user preferences influenced by sentiments compared to the other similarity measures evaluated.

However, it is important to note that while the inclusion of sentiment analysis resulted in a reduction in RMSE from 1.2703 (without sentiment analysis) to 1.2478 (with sentiment analysis) for cosine similarity, the improvement was not statistically significant. A t-test performed on the RMSE values revealed a T-statistic of -0.3485 and a p -value of 0.7450 , indicating that the observed change in RMSE is not statistically significant. This suggests that while sentiment analysis may offer some improvement in prediction accuracy, the difference in performance may not be substantial enough to be considered statistically meaningful.

As we move forward, it is crucial to explore more advanced approaches, such as hybrid models that combine collaborative filtering with content-based methods using feature combinations, to potentially achieve more significant gains in recommendation accuracy.

4.5 Performance comparison of hybrid models

In our study, we explored a variety of hybrid models that combined cosine prediction for collaborative filtering, clustering for content-based filtering, and BERT prediction for sentiment analysis. To evaluate the performance of these models, we utilized RMSE as a measure of accuracy. By comparing the RMSE values of the hybrid models with those of traditional approaches, we assessed their effectiveness in providing accurate recommendations. The results revealed that the hybrid models consistently outperformed the traditional methods, achieving lower RMSE scores. This indicates that the hybrid models were more adept at capturing user preferences, item characteristics, and sentiment information, resulting in more precise and personalized recommendations.

Table 10 shows the performance of different recommendation models based on their RMSE values. The models evaluated include the Weighted Hybrid, RandomForestRegressor, NMF + RandomForestRegressor, GradientBoostingRegressor, NMF + GradientBoostingRegressor, SVR, NMF + SVR, DecisionTreeRegressor, and NMF + DecisionTreeRegressor. The RMSE values indicate the average difference between the predicted and actual ratings, with lower values indicating higher accuracy. Among the models, the NMF + DecisionTreeRegressor hybrid model achieved the lowest RMSE of 1.1955, demonstrating its superior performance in generating precise and reliable recommendations.

The lower RMSE of our proposed NMF + DecisionTreeRegressor hybrid model (1.1955) compared to others, such as NMF + RandomForestRegressor and NMF + GradientBoostingRegressor, can be attributed to several factors:

1. *Feature combination*: The integration of collaborative filtering, content-based filtering, and sentiment analysis into a unified feature set allows our model to capture diverse aspects of user preferences and item characteristics more comprehensively. This approach leads to more precise predictions.
2. *NMF*: Nonnegative matrix factorization (NMF) effectively decomposes the user–item matrix into latent factors, capturing hidden patterns in user preferences and item attributes. This decomposition enhances the model’s ability to identify relevant recommendations.
3. *DecisionTreeRegressor*: The DecisionTreeRegressor, with its ability to handle complex interactions and nonlinear relationships between features, refines the recommendations further based on the comprehensive feature set provided by NMF.

In contrast, models like NMF + RandomForestRegressor and NMF + GradientBoostingRegressor, while robust, may not capture the specific nuances of user preferences and sentiment as effectively due to their different feature handling and learning strategies.

Overall, our hybrid model’s lower RMSE reflects its enhanced capability to deliver accurate and personalized recommendations by leveraging a combination of sophisticated techniques in feature extraction and regression.

4.6 Final recommendation system

Table 11 shows a comparative analysis of different recommendation systems (RS) and clustering approaches, shedding light on their performance metrics. Collaborative filtering (CF), a widely used method, yielded an RMSE value of 1.2703, providing an insight into the accuracy of predicting user preferences. However, when sentiment analysis (SA) was integrated into the CF approach, the RMSE improved to 1.2478, suggesting that considering SA could enhance the precision of rating predictions. Content-based (CB) clustering, on the other hand, achieved a silhouette score of 0.1922, indicating the quality of the formed clusters. Nevertheless, the incorporation of SA into the CB clustering approach resulted in a higher silhouette score of 0.2270, indicating improved clustering performance. Moreover, our model, which incorporates both sentiment analysis and advanced techniques, achieved outstanding results with an RMSE of 1.1955. This emphasizes the potential of SA in improving RS. In our study, we utilized a hybrid model that combined nonnegative matrix factorization (NMF) with the DecisionTreeRegressor. This approach allowed us to leverage the strengths of both techniques and improve the accuracy and performance of our RS. By integrating NMF and DecisionTreeRegressor, we achieved more accurate recommendations by considering both user–item interactions and item characteristics.

While the improvement in RMSE for the collaborative filtering model after integrating sentiment analysis was found to be statistically insignificant (p -value = 0.7450), our hybrid

Table 11 Performance evaluation of collaborative filtering, content-based, and our model with sentiment analysis

Recommendation Model	Evaluation Metric	Score
Collaborative Filtering	RMSE	1.2703
Collaborative Filtering + Sentiment	RMSE	1.2478
Content-Based with Clustering	Silhouette score	0.1922
Content-Based with Clustering + Sentiment	Silhouette score	0.2270
Our model	RMSE	1.1955

model, which combines collaborative filtering and content-based filtering using a feature combination approach (NMF + DecisionTreeRegressor), achieved a notable reduction in RMSE (1.1955). This demonstrates the effectiveness of sentiment analysis when incorporated into a hybrid recommendation system, highlighting the practical value of the proposed approach in improving prediction accuracy.

4.6.1 Recommendation system error analysis

While our model demonstrated superior performance compared to other approaches, it is important to recognize and understand the scenarios where it did not perform optimally. Common sources of error include the “cold start” problem, where insufficient data for new users or items leads to less accurate predictions. Additionally, users with rare or highly specific preferences posed challenges for the model, resulting in less relevant recommendations. The parameters used in the DecisionTreeRegressor also played a critical role in the model’s performance. Suboptimal parameter settings could lead to overfitting or underfitting, thereby reducing the accuracy of predictions in certain cases.

Understanding these errors is crucial for improving the recommendation system. Future work could focus on refining the model parameters, integrating additional data sources, or employing hybrid methods to address these limitations and enhance the overall accuracy and robustness of the system.

4.6.2 Comparison with existing research

In comparing our approach to previous works, as summarized in Table 12, it is evident that our integration of sentiment analysis using BERT yields a notable accuracy of 91%, surpassing Work [44]’s XGBoost method, which achieves an accuracy of 89%. This advantage in sentiment accuracy is significant as it reflects the robustness of BERT in understanding the nuances of user sentiments. Additionally, our use of a hybrid recommendation system that combines both content-based and collaborative filtering methodologies demonstrates a lower RMSE of 1.1955, indicating more precise predictions compared to Work [44], which reports an RMSE of 1.36 when employing a correlation similarity approach. Furthermore, our content-based clustering using K-means achieved a silhouette score of 0.2270 with 7 clusters, determined using the elbow method, outperforming Work [44]’s score of 0.193. Work [45], while utilizing a collaborative filtering method based on SVD, yields a higher RMSE of 1.3, further establishing the effectiveness of our proposed hybrid system. Overall, these comparisons suggest that our methodology not only provides enhanced sentiment analysis

Table 12 Comparison of our approach with existing research on hybrid recommendation systems using sentiment analysis

Approach	Sentiment analysis method	Sentiment analysis (accuracy)	Content-based	Collaborative filtering	Hybrid recommendation system (RMSE)
Our work	BERT	91%	K-means (7 Clusters, determined using Elbow Method), Silhouette Score: 0.2270	Cosine Similarity, RMSE: 1.24	1.1955
Work [44]	XGBoost	89%	K-means (250 Clusters, Silhouette Score: 0.193)	Correlation Similarity, RMSE: 1.36	N/A
Work [45]	N/A	N/A	N/A	SVD, RMSE: 1.3	N/A

but also integrates these insights more effectively into the recommendation process, leading to improved accuracy in user preference predictions.

5 Conclusion and future work

In conclusion, the work on integrating sentiment analysis (SA) into a recommendation system (RS) has yielded promising results. This study employed BERT for SA, cosine similarity for collaborative filtering (CF) with SA, and CB clustering incorporating sentiment predicted by BERT. These techniques were combined using NMF with DecisionTreeRegressor, resulting in a recommendation model with an improved RMSE of **1.1955**.

The individual components of the system demonstrated their effectiveness. The SA using BERT achieved an impressive accuracy rate of **91%**, enabling the system to accurately capture and incorporate user sentiments into the recommendation process. CF combined with SA yielded a reduced RMSE of 1.2478, indicating improved accuracy by considering user sentiments in the recommendation process.

Moreover, the CB approach with clustering, incorporating sentiment predicted by BERT, achieved a higher silhouette score of 0.2270 compared to 0.1922 without SA. This suggests that incorporating sentiment information enhances the clustering performance and improves the quality of CB recommendations.

While the improvement in RMSE for the collaborative filtering model after integrating sentiment analysis was found to be statistically insignificant (p -value = 0.7450), our hybrid model, which combines collaborative filtering and content-based filtering using a feature combination approach (NMF + DecisionTreeRegressor), achieved a notable reduction in RMSE (1.1955).

Our proposed model, which combines the strengths of CF, CB clustering, and SA, outperformed the individual techniques and achieved a more accurate RS with an RMSE of 1.1955. This indicates that the model effectively leverages the advantages of each approach to provide personalized and sentiment-aware recommendations.

In conclusion, the integration of SA into RS, as demonstrated in this work, shows promise for enhancing recommendation accuracy and personalization. Continued research and refinement of the proposed model can contribute to the advancement of sentiment-aware RS, benefiting users in various domains. Future research could focus on exploring alternative sentiment analysis models, such as RoBERTa approaches, to improve sentiment classification accuracy. Additionally, investigating advanced fusion techniques and incorporating diverse data sources, like social media interactions or user demographics, could further enhance the recommendation system's effectiveness.

Data Availability The datasets used and analyzed during the current study are publicly available and can be accessed at [37]. Data sharing is applicable to this article as datasets were generated and analyzed during the study.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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